Md Faiz Akhtar

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Available to start immediately

Research Summary

Computational materials physicist working on first-principles electronic structure and AI for materials physics. PhD work uses DFT electronic-structure and bonding descriptors to predict macroscopic behaviour (slip-system selection, interstitial strengthening) in titanium and HCP metals, carried up through MD and crystal-plasticity FE. Develops machine-learned interatomic potentials (MACE fine-tuning with GPU committee active learning on a 1600+ VASP database) and *ICME Agent*, an autonomous ∼55-agent platform that runs, validates and interprets VASP, LAMMPS and DAMASK workflows on HPC with deterministic, regression-tested physics. Authors condensed-matter and semiconductor-transport problems for frontier-AI scientific benchmarks (SciCode v2, Agentic SciCode).

Education

Ph.D., Materials Science & Engineering, Indian Institute of Technology Kanpur 2021 – present | CPI: 8.5/10
Thesis: Computational investigation of titanium, bridging length scales from electronic to continuum
Advisors: Prof. Somnath Bhowmick and Prof. Nilesh P. Gurao

M.Tech., Materials Science & Engineering, Indian Institute of Technology Jodhpur 2019 – 2021 | CPI: 9.03/10
Thesis: First-principles screening of Ti_2C-based MXenes as electrode materials for Li-ion batteries

B.E., Mechanical Engineering, Institute of Technology Korba, Chhattisgarh 2013 – 2017 | 76.67%

Class XII (CGBSE), Shiva Public School, Bhilai 2013 | 79%

Fellowships & Awards

- **Prime Minister's Research Fellowship (PMRF)**, Government of India, the nation's most prestigious doctoral fellowship. Aug 2022 – Jul 2025
- **GATE (Mechanical Engineering)**, qualified the national Graduate Aptitude Test in Engineering. 2019

Publications

Journal Articles

1. **Md Faiz Akhtar**, et al., "A novel electronic structure-based prediction of interstitial strengthening in α-titanium," *Journal of Applied Physics* **138**, 175111 (2025). [link]
2. **Md Faiz Akhtar**, et al., "Predicting primary slip systems in HCP metals: a combined geometrical and electronic approach," *Materials Letters* **400**, 139019 (2025). DOI: 10.1016/j.matlet.2025.139019
3. U. Rajput, **Md Faiz Akhtar**, et al., "Evolution of atomistic boron clusters on borophene monolayers on Au(111)," *Journal of Physical Chemistry C* (2025). DOI: 10.1021/acs.jpcc.5c00397
4. A. Linda, **Md Faiz Akhtar**, et al., "Accelerating generalized stacking-fault energy prediction by combining the Friedel model, ab initio calculation and machine learning," *Physical Review B* **109**, 214102 (2024). DOI: 10.1103/PhysRevB.109.214102
5. U. Rajput, **Md Faiz Akhtar**, et al., "Liquid-phase exfoliated borophene on Au(111)," *Journal of Physical Chemistry C* (2024). DOI: 10.1021/acs.jpcc.4c00497
6. **Md Faiz Akhtar**, et al., "Linking electronic bonding and short-range order to strength in α-titanium alloys: a first-principles study," *Materials Chemistry and Physics* (2026). [link]

Book Chapters

7. A. Linda, **Md Faiz Akhtar**, et al., "Deformation in metals: insights from ab-initio calculations," Springer (2024). DOI: 10.1007/978-981-99-6863-3_10

In Preparation

8. **Md Faiz Akhtar**, et al., "Direction-resolved bond anisotropy predicts slip-system selection in HCP metals and alloys."
9. **Md Faiz Akhtar**, et al., "An autonomous multi-scale ICME agent framework: seamless integration of ab initio DFT, molecular dynamics and crystal-plasticity FE via automated scale-bridging."
10. **Md Faiz Akhtar**, N. P. Gurao, S. Bhowmick, "An auditable agent that chooses multiscale calculations for titanium alloys from the literature."
11. **Md Faiz Akhtar**, et al., "COMPASS: a hand-verified, cross-database resource for binary-alloy phase behaviour, and which descriptors (and which DFT) actually matter."
12. **Md Faiz Akhtar**, et al., "Fine-tuning a multi-head MACE foundation model for mechanical properties of d-block high-entropy alloys: benchmark, DFT anchor database and committee active learning."

13. **Md Faiz Akhtar**, et al., "MD-informed anisotropic phase-field modeling of grain growth in polycrystalline copper."
14. **Md Faiz Akhtar**, et al., "Computational investigation of titanium: bridging length scales from electronic to continuum."
15. S. Mahato, **Md Faiz Akhtar**, et al., "Inverse grain size dependence of strain rate sensitivity in a HPT-processed $(CrFeNi)_{99}Si$ medium entropy alloy."
16. **Md Faiz Akhtar**, et al., "Functionalized Ti_2C-based MXenes for Li-ion battery applications."

Research Projects & Software

Fine-Tuned MACE Potentials for High-Entropy Alloys, Committee Active Learning on GPUs Ongoing

- Fine-tuning the MACE-MH-1 foundation model for the mechanical response of fcc Cantor-type and bcc refractory high-entropy alloys, with CoCrFeMnNi and VNbMoTaW held out as transferability tests; zero-shot benchmark of seven MACE foundation models on 307 d-block crystals (best: 9.2% median shear-modulus error).
- Built a spin-polarised PBE anchor database of over 1600 VASP calculations (elastic strains, vacancies, stacking faults, surfaces, ideal strength paths) via Slurm task farms on the PARAM Rudra supercomputer.
- Trained ten-member committees on 10 GPUs over successive active-learning rounds, cutting validation errors from 40.8 meV/atom and 84.9 meV/Å (zero-shot) to 2.4 meV/atom and 38.5 meV/Å; reproduced DFT lattice parameters to 0.002 Å and stacking-fault and surface energies to 5.5%.

ICME Agent, Autonomous Multi-Agent AI System for Multi-Scale Materials Simulation Ongoing
akhtar07.github.io/atoms-to-agents | *Advisors: Prof. Somnath Bhowmick and Prof. Nilesh P. Gurao*

- Architected a multi-agent AI platform of ∼55 cooperating Python agents (∼68k LOC) that turns plain-English materials-science requests into fully executed multi-scale workflows, autonomously planning, submitting, monitoring, validating, and post-processing VASP (DFT), LAMMPS (MD), and DAMASK (crystal-plasticity FE) simulations across a remote SLURM HPC cluster and local compute.
- Engineered a deterministic-core / LLM-narration design in which physics correctness never depends on a language model: a fully local LLM (Qwen2.5-32B via Ollama) only parses intent and narrates results, while all correctness-critical numbers come from validated deterministic code, eliminating hallucination risk in scientific output at zero paid-API cost.
- Automated end-to-end DFT workflows (input generation, convergence testing, wall-time-aware SLURM dispatch, error recovery) and a DFT→DAMASK coupler feeding first-principles elastic constants into mesoscale plasticity; validated against experiment, e.g. Al monovacancy formation energy 0.681 eV vs. ∼0.67 eV experimental.
- Implemented a physics library spanning derived mechanical properties (Debye temperature, sound velocities, Pugh ratio, Vickers hardness), quasi-harmonic temperature-dependent thermodynamics and elasticity (phonopy + Debye–Grüneisen), and electronic/optical property extraction, validated against literature, e.g. tungsten thermal expansion 4.18 vs. 4.5 ppm/K and sound velocities within 0.3% of experiment.
- Built a closed-loop discovery agent (SciAgents-style ideation over a materials knowledge graph → executable simulations → deterministic physics critic → automatic refinement) that self-corrects failure modes without human intervention, exposed through a real-time natural-language web GUI (FastAPI + WebSockets).
- Ensured rigor with 26 regression test suites validating physics against analytic and known values, plus real multi-hour HPC campaigns; currently integrating MACE machine-learning interatomic potentials (GPU-accelerated) to extend all workflows across the periodic table via a single backend swap.

TiSci, an Auditable Literature-to-Calculation Agent for Titanium Alloys 2026
akhtar07.github.io/tisci-explorer | *Advisors: Prof. Somnath Bhowmick and Prof. Nilesh P. Gurao*

- Built an agent that screens 203,063 titanium works into a knowledge graph (1.35M nodes, 3.36M edges; local Qwen3 LLMs served with vLLM), generates and tournament-ranks hypotheses with critic, data-checker and novelty roles, and turns each into one calculation chosen by value per cost or expected information gain.
- Verdicts are computed in code from sign checks replicated under two protocols (MACE-MP-0), and accepted results are carried without human choice through ICME-lite (DFT γ-lines → Peierls–Nabarro CRSS → DAMASK and VPSC crystal plasticity for fcc, bcc and hcp metals); every number is regenerable from stored records.
- Ran a 283-cycle, 228 h autonomous campaign mapping 16 solutes, 120 pairs and 455 triples in α-Ti, and calibrated the hand-built DFT-to-crystal-plasticity chain against tension tests and EBSD at 298–143 K (flow-curve RMSE of 15.5 MPa in DAMASK and 6.8 MPa in VPSC); the audit trail exposed the agent's own failure modes.

COMPASS Explorer, Cross-Database Resource for Binary-Alloy Phase Behaviour Ongoing
akhtar07.github.io/compass-explorer

- Built a hand-verified, cross-database resource reconciling binary-alloy phase-behaviour data across major materials databases, exposed through an interactive web explorer.
- Benchmarking which thermodynamic and electronic-structure descriptors, and which levels of DFT, actually matter for predicting binary-alloy phase stability.

Earlier Course Projects, IIT Kanpur 2021

- Determination of dihedral angles at grain-boundary triple junctions via image processing (with Prof. Sandeep Sangal), Python pipeline (scikit-image, NumPy) locating grain boundaries and triple junctions in polycrystalline micrographs.
- First-principles assessment of borophene as a Na-ion battery anode (with Prof. Somnath Bhowmick), DFT study of structure, electronic properties, and energy-storage capacity.

Professional Experience

Scientific Task Author, Imperium | Sci-Agent, Mercor (Contract) Jul 2026 – present

- Authoring frontier agentic-evaluation tasks that require LLM agents to autonomously plan, execute, and verify complete multi-step scientific research workflows, from literature comprehension to quantitative verification of results.
- Extending the Agentic SciCode evaluation methodology (deterministic oracle solutions, programmatic + LLM-as-judge grading, containerized runtimes) to open-ended scientific agent evaluation.

Scientific Benchmark Author, SciCode v2, Turing (Contract) May 2026 – present

- Authoring problems for SciCode v2, a benchmark evaluating the scientific reasoning and coding ability of frontier LLMs (GPT-5.5, Gemini 3.1 Pro, Claude Opus 4.6) on hard problems drawn directly from published research in physics and materials science.
- Translate primary literature (condensed-matter physics, semiconductor transport, materials science) into self-contained, LaTeX-formatted coding problems demanding non-trivial physics from first principles.
- Author golden reference solutions and discriminative test suites (3–4 unit tests per sub-problem; tolerances calibrated from rtol 10^{-4} to 10^{-9}) that separate genuinely correct implementations from plausible-but-wrong physics.
- Audit and rewrite auto-generated notebooks, removing scientifically invalid patterns (post-hoc output clamping, hard-coded trend enforcement) and replacing them with faithful implementations derived from the source paper.
- Calibrate problem difficulty through iterative RLHF (4-run) and Pass@K (8-run) evaluation cycles to hit target pass-rate bands (0–40% main problems, 10–75% sub-problems) with strict validator compliance.

AI Benchmark Task Author, Agentic SciCode (Project Imperium), Mercor (Contract) Apr 2026 – Jun 2026

- Authored research-grade tasks for a 260-task frontier agentic benchmark assessing whether LLM agents (GPT-5, Claude Sonnet 4.6) can reason and execute like scientists; tasks spanned error diagnosis, end-to-end analysis, code generation, and result interpretation in materials science and physics simulation.
- Sourced tasks from post-cutoff (post-Aug 2025) arXiv papers to guarantee out-of-distribution reasoning, calibrated to Level-3 difficulty (35–65% frontier-model pass rate) under a multi-label capability taxonomy.
- Wrote expert golden solutions and deterministic Python oracles (seeded RNGs, configuration-driven, fully reproducible) producing the complete expected output set: code, figures, structured reports, and `answer.json`.
- Engineered dual-component evaluation harnesses per task: programmatic verifiers (numerical checks against golden values, schema validation, structural gates; 30% weight) and a multimodal LLM-as-judge scoring 25+ rubric criteria on scientific reasoning, completeness, and presentation (70% weight).
- Built Dockerized agent environments with domain-specific scientific stacks (NumPy, SciPy, ASE, PyMatGen, RDKit, PyTorch) and iterated tasks through a multi-gate CI/QC pipeline (oracle validation, central QC, dual frontier-model calibration) using GitHub Actions.

AI Research Contributor, Human Frontier Collective, Scale AI May 2025 – Oct 2025

- Selected for Scale AI's Human Frontier Collective, an invite-based program of Ph.D.-level researchers supporting frontier generative-AI development.
- Contributed expert-level evaluation, red-teaming, and critical analysis of next-generation proprietary models.

Conference Presentations

- "Effect of interstitial atoms on the deformation behavior of titanium: an ab-initio study," *International Symposium on Transformational Technologies in Materials & Manufacturing*, UAS Bengaluru, Nov 2024.
- "Computational investigation of titanium: bridging length scales from electronic to continuum," *30th International Conference on Computational & Experimental Engineering and Sciences (ICCES)*, Singapore, Aug 2024.
- "Computational investigation of titanium: bridging length scales from electronic to continuum," *HPC Symposium 2024*, IIT Kanpur, Apr 2024.
- "Accelerating the prediction of stacking fault energy by combining ab initio calculations and machine learning," *PMRF Symposium 2024*, IIT Indore, Mar 2024.
- "Predicting primary slip systems in HCP materials: a combined geometrical and electronic approach," *IIM-ATM 2023*, KIIT University, Bhubaneswar, Nov 2023.
- "A critical role of oxygen on the deformation behaviour of Ti and Ti–Al alloys," *METCENT 2023*, IIT (BHU) Varanasi, Oct 2023.

Workshops & Schools

- **LOBSTER School on Chemical Bonding Analysis**, BAM Akademie, Berlin, Germany. Jun 2025
- **GIAN Course: Modelling Defects in Crystalline Solids**, Prof. David Rodney, IIT Madras. Jan 2025

Technical Skills

Electronic structure & atomistic simulation: VASP, Quantum ESPRESSO, Phonopy (quasi-harmonic phonons), QuantumATK, LAMMPS, DAMASK, PFDD, CALPHAD, atomicrex, LOBSTER (chemical bonding analysis), MACE (ML interatomic potential fine-tuning, committee active learning)
Programming & HPC: Python (FastAPI, WebSockets, asyncio), Bash, LaTeX; Linux/HPC clusters, SLURM, multi-GPU model training (PyTorch)
Scientific Python: NumPy, SciPy, scikit-learn, PyTorch, Matplotlib, pymatgen, ASE, OVITO Python API, SQLite3, Paramiko, PyYAML
AI & LLM engineering: multi-agent system architecture, closed-loop autonomous agents, knowledge graphs and RAG, LLM benchmark design and evaluation, LLM-as-judge rubric engineering, pass-rate calibration, Anthropic Claude API, local LLM deployment (Ollama, Qwen2.5-32B)
DevOps & tooling: Docker, Git, GitHub Actions, JSON Schema, TOML
Visualization: VESTA, OVITO, Xmgrace, gnuplot

Academic Service & Leadership

- **Overall Ph.D. Placement Coordinator**, Students' Placement Office, IIT Kanpur, leading placement activities for the institute-wide Ph.D. cohort. 2025 – 26
- **Senator**, Students' Senate, IIT Kanpur, elected representative of Ph.D. students in institute policy-making.
- **President**, Indian Institute of Metals (IIM) Kanpur Chapter, organized quizzes, workshops, and lectures.
- **Overall Coordinator**, Materials Science Society (MatSoc), IIT Kanpur, organized departmental events.
- **Student Secretary**, National Symposium of Research Scholars on Metallurgy and Materials (NSRS-2024).
- **Institute Badminton Team**, IIT Kanpur, represented the institute at UDGHOSH 2022, 2023, and 2024.